UNIVERSITY of the WESTERN CAPE
UNIVERSITEIT VAN PRETORIA UNIVERSITY OF PRETORIA YUNIBESITHI YA PRETORIA
UNIVERSITY OF MPUMALANGA
LUANAR
UNIVERSITY OF HOHENHEIM

UKUDLA Data Science Certificate

Introduction to Data Science for Food Systems Analysis

Markus Mößler

University of Hohenheim

UKUDLA
African German Centre
for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

science, technology & innovation
Department: Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA

NRF National Research Foundation

Supported by:
Federal Foreign Office

With funding from the:
Federal Ministry of Research, Technology and Space

With support from
Federal Ministry of Agriculture, Food and Regional Identity

by decision of the German Bundestag

DAAD Deutscher Akademischer Austauschdienst
German Academic Exchange Service

Table of contents

UKUDLA

Why data science matters for food systems

Food systems generate diverse evidence

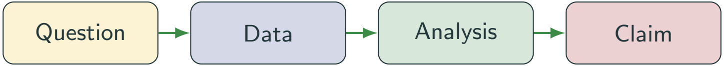
- laboratory measurements and assays;
- designed field experiments;
- field observations, surveys and qualitative evidence; and
- administrative, market, climate and remote-sensing data.

Decisions require more than data

- transparent questions and assumptions;
- appropriate preparation and analysis;
- reproducible computational workflows; and
- careful interpretation of uncertainty.

Data science connects questions, data, computation and judgment to produce evidence that can be inspected and reused.

The challenge is a chain of decisions



- Available data may not represent the intended population or concept.
- Transformations and models encode assumptions that affect conclusions.
- Results can be technically correct but scientifically misleading.
- Documentation and reproducibility make the decision chain reviewable.

Data science is an integrated practice

Domain knowledge
Defines meaningful questions, measurements and interpretations.

Statistical reasoning
Describes variation, evaluates relationships and quantifies uncertainty.

Computing practice
Acquires, transforms, analyzes and communicates data reproducibly.

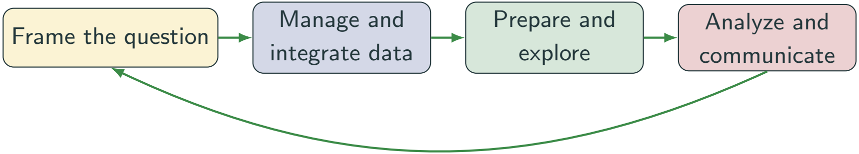
No single tool defines data science. Credible work depends on how these forms of knowledge are combined.

Different questions require different analyses

Purpose	Question	Example output
Descriptive	What patterns are present in the observed data?	Summaries, distributions and trends
Explanatory	How should an assumed relationship be estimated and interpreted?	Effect estimate with assumptions
Predictive	How accurately can an unknown outcome be predicted?	Out-of-sample predictions and error

- A useful analysis begins with a clearly stated purpose.
- Explanatory and predictive success are not interchangeable.
- Every conclusion is conditional on data quality and design choices.

A data science workflow is iterative



- Findings often reveal new data-quality issues or better questions.
- Iteration is expected; undocumented iteration is the risk.
- Versioned inputs, code and decisions preserve the path to a result.

Credibility requires reproducibility and responsibility

Reproducible practice

- version code and documentation;
- record software environments;
- automate transformations; and
- preserve provenance and validation.

Responsible practice

- respect licenses, privacy and consent;
- examine representation and bias;
- distinguish evidence from assumptions; and
- communicate limitations honestly.

A result is credible only when its production and interpretation can be examined.

Part 1 builds tools for reproducible work

Version control
Record meaningful project states and collaborate through Git and GitHub.

Environments
Recreate package and system dependencies with `renv` and Docker.

Remote computing
Work safely on other computers through SSH and Linux tools.

These tools support every later topic; they are part of the analytical method, not administrative extras.

Part 2 establishes trustworthy analytical data

Data management
Describe, organize, validate and preserve data and their history.

Data integration
Align identifiers, time, space and meaning across sources.

Data preparation
Transform managed data into an explicit unit of analysis.

Analytical validity begins before modeling: later results inherit the choices made in constructing the data.

Part 3 develops complementary forms of analysis

Topic	Primary role
Data visualization	Reveal structure and communicate evidence through graphical encodings
Descriptive analysis	Summarize distributions, dependence, change and stability
Explanatory analysis	Estimate and qualify relationships under causal assumptions
Predictive analysis	Evaluate forecasts on observations not used for model fitting

- Each topic combines motivation, reusable concepts and application guidance.
- Detailed pages support study; the example repository supports practice.

Key message: make the complete reasoning chain visible

Data science is the disciplined construction of evidence from questions, data, computation and assumptions—not merely the use of software or models.

- Start with the purpose of the analysis.
- Treat data and computational choices as part of the evidence.
- Match descriptive, explanatory or predictive methods to the question.
- Preserve enough information for others to inspect, reproduce and reuse the work.

Supported by

South African government and the National Research Foundation

science, technology & innovation

Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA

NRF National Research Foundation

German government and the German Academic Exchange Service

Supported by:
Federal Foreign Office

With funding from the:
Federal Ministry of Research, Technology and Space

Supported by:
Federal Ministry of Agriculture, Food and Regional Identity

by decision of the German Bundestag

Project Manager:
Federal Office for Agriculture and Food

DAAD
Deutscher Akademischer Austauschdienst
German Academic Exchange Service